



PROJECT:

Network Design for Faculty of Computing Block

TASK 3: CHOOSING THE APPROPRIATE LAN DEVICES

SECR1213 - NETWORK COMMUNICATIONS

SEMESTER I, SESSION 2024/2025

SECTION 02

Lecturer: Dr. ISMAIL FAUZI BIN ISNIN

Name	Matric Number
TAN ZHI MING	A23CS0189
NG YU HIN	A23CS0148
LEE YIN SHEN	A23CS0236
MUHAMMAD NAIM BIN ABDULLAH	A23CS0134

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3.1 Introduction of Required Devices

Router

The router connects to multiple networks and guides packets between them, ensuring efficient communication. They manage traffic between the internal network and the Internet, assign IP addresses, and implement routing protocols. Advanced routers also provide features such as quality of service (QoS), VPN support, and secure filtering to ensure optimized and secure data flow.

Switches

Switches are essential for internal network connectivity, enabling devices such as computers, printers, and servers to communicate within the same local area network (LAN). They use MAC addresses to forward data only to designated devices, thereby improving network efficiency and reducing traffic.

Firewalls

Firewalls act as a security barrier between trusted internal networks and untrusted external networks, monitoring and controlling incoming and outgoing traffic. They use predefined rules to block unauthorized access and protect against threats such as malware, hackers, and DDoS attacks, ensuring the integrity of sensitive data and systems.

Access Points (Wi-Fi)

Access points extend wired networks wirelessly, allowing devices such as laptops, smartphones and tablets to connect to the network. Modern access points support advanced wireless standards such as Wi-Fi 6 for high-speed, reliable connectivity, and include features such as load balancing, customer networking, and security protocols such as WPA3.

Network Attached Storage (NAS)

NAS provides centralized shared storage for files and data on a network. It allows multiple users and devices to access data simultaneously, and supports file sharing, backup, and media streaming. NAS devices typically include RAID for redundancy and security features for data protection.

Servers

A server is a powerful computer that hosts and manages resources, services, and applications for the network. They handle tasks such as authentication (for example, Active Directory), file storage, database hosting, and virtualization. The server ensures high availability and centralized control of critical network operations.

IoT Gateway

IoT gateways connect IoT devices and networks to enable secure communication between sensors, controllers and the cloud. They translate protocols, pre-process data for analysis, and ensure seamless connectivity while providing security features to protect IoT deployments from cyber threats.

Ethernet Cables (Cat6a)

Ethernet cables connect devices such as computers, switches and routers in the local network, transmitting data at speeds of up to 10Gbps over distances of 100 meters. Cat6a cables are shielded to reduce interference and ensure stable, high-speed connections for demanding applications such as video streaming and large file transfers.

Fiber Optic Cables

Fiber optic cables use light to transmit data, provide ultra-high speeds, low latency, and support long distances (up to several kilometres). They are ideal for backbone connections between switches, server rooms, and critical areas, ensuring reliable performance for high-bandwidth tasks.

Patch Panels

The distribution board organizes network cables in a centralized manner, providing a clean and efficient way to connect devices to switches or other network devices. They allow easy access to a single port for reconfiguration or testing, simplifying troubleshooting, maintenance, and upgrades.

Keystone Jacks

Keystone jacks are modular connectors used to terminate Ethernet cables in wall sockets or distribution boards. They provide a secure, standardized interface for the insertion of network devices, ensuring stable connections and minimizing signal loss.

RJ45 Connectors

The RJ45 connector is a standardized terminal for Ethernet cables that can be used to connect network devices such as computers, switches and routers. They are easy to install and support high-speed data transfer over Cat6a cables.

Conduits and Trunking

Conduits and mains are the shells that protect network cables from physical damage, environmental interference, and electromagnetic interference. They also ensure neat cable management, making installations more durable, organized, and visually appealing.

Personal Computer (PC)

High-performance PCs are essential for both labs classrooms like General Labs, Cisco Networking Lab, Embedded Systems Lab and Hybrid Classroom in order to meet the requirements for advanced academic activities. A PC provides the student and staff with primary interface for network access, software applications, and practical learning tasks. By equipping the labs and classroom with powerful and reliable PCs, the faculty can ensure smooth multitasking and high-speed connectivity for both teaching and practical tasks, while fulfilling the growing needs of modern education.

Smart TV

Smart TVs will be set up in the Cisco Networking Lab, Embedded Systems Lab and Hybrid Classroom to enhance the teaching and learning experiences. With high-resolution display and advanced features, a smart TV allow an educator to deliver interactive lessons like demonstrating a real-time application. This device ensures the academic environment stays at the forefront of educational technology while supporting collaborative and hybrid learning environment.

Projector

Projector will be installed in the General Labs and Meeting Room to provide large-scale visual displays of lectures. In the labs, a projector is important to showcase hand-on education contents like code walkthroughs or live demonstrations while in the meeting room, a projector used to display slides, documents and multimedia to support communication and decision-making.

Speaker

Speaker is important to amplify the audio output in labs, hybrid classroom and meeting room, which ensure the students and staffs can clearly hear an audio during lectures or presentations. A high-end speaker will enhance the delivery of instructional content and support a more engaging and interactive learning environment.

Microphone

Microphone will be prepared in both labs and classrooms. A microphone can capture clear audio for lectures, discussions and hybrid learning sessions, lead to effective communication between staff and students. By pairing to Smart TV, projector and speaker, microphone ensure that the voices of

educators transferred clearly, which is able to improve overall efficiency of teaching and collaboration processes.

3.2 Table of Devices

Device	Specification	Price per unit	Quantity	Cost
Core Device				
Routers 	<u>Cisco ISR 4000 Series/k90</u> - Concurrent software services at speeds up to 10 Gbps (ISR4461). Backplane architecture supports high-bandwidth module-to-module communication at speeds up to 10 Gbps. - Embedded Cisco Catalyst SD-WAN solution for creating lower-cost, business-class Internet connections - Remote installation of application-aware services, which run identically to their counterparts in dedicated appliances.	5042.29	2	10084.58
Switches 	<u>C9300l-48t-4x-e, 48-port 1G copper with fixed 4x10G/1G SFP+ uplinks, data only, Network Essentials</u> - With Stackwise-1T, Catalyst 9300 switches include high-density stacking bandwidth solution with the most flexible uplink architecture. - Flexible and dense uplink offerings with 100G, 40G, 25G, Multigigabit, 10G, and 1G modular uplinks.	9307.78	5	46538.88

	48 mGig ports in standalone and 448 mGig ports with an 8-member stack.			
Firewalls	<u>Cisco fpr2110-ngfw-k9</u> - Cisco Firepower NGIPS available which can passively detect endpoints and infrastructure for threat correlation and Indicators of Compromise (IoC) intelligence - Cisco AMP for Networks, available enables detection, blocking, tracking, analysis, and containment of targeted and persistent malware, addressing the attack continuum both during and after attacks. Integrated threat correlation with Cisco Secure Endpoint is also optionally available	12500	1	12500
Access Points (Wi-Fi) 	<u>C9130AXI-A</u> 8x8 uplink/downlink MU-MIMO with eight spatial streams 802.11ax beamforming 20-, 40-, 80-, and 160-MHz channels	2750	16	44000

	- PHY data rates up to 5.38 Gbps (8x8 80 MHz or Dual 4x4 80+80 MHz on 5GHz and 4x4 20 MHz on 2.4) - CSD support - WPA3 support			
NAS (Network Attached Storage) 	<u>Cisco UCS S3260</u> - Dual two-socket server nodes based on Intel® Xeon® Scalable processors or Intel E5-2600 v4 CPUs with up to 44 cores per server node - Up to 1.5 TB of DDR4 memory per server node (3 TB total) - Massive 960 TB data storage capacity that easily scales to Petabytes with UCS Manager - Dual-port 40 Gbps System I/O Controllers with Cisco VIC 1300 Series Embedded Chip	67608.59	1	67608.59
Servers 	<u>Cisco UCS C480M5</u> - Memory speeds increase from 2666 MHz to 2933 MHz - Up to 4 GPUs (2-CPU configuration) or 6 GPUs (4-CPU configuration) with offerings from NVIDIA and AMD - Support for higher-density DDR4 memory, from 6 TB	50000	3	150000

	(128G DDR4 DIMMs) to 12 TB (256G DDR4 DIMMs)			
IoT Gateway 	<u>Raspberry Pi 4 IoT Gateway</u> - Gigabit Ethernet for high-speed wired connectivity. - Two USB 3.0 ports for high-speed devices. - Two USB 2.0 ports for compatibility with other peripherals. - Operating Systems: Runs Raspberry Pi OS (Linux-based) and other OS options like Ubuntu, or specialized IoT platforms like BalenaOS.	300	3	900
Power & Cooling System				
Uninterruptible Power Supply (UPS) 2 server room 2 database room 1 VCR 1 IOT 	<u>APC Smart-UPS, Line Interactive, 1500VA, Rackmount 2U, 120V, 6x NEMA 5-15R outlets, SmartConnect Port+SmartSlot, AVR, LCD</u> 1440VA/1000W line interactive pure sine wave 2U rackmount UPS - Automatic Voltage Regulation helps correct utility voltage - Cloud-based remote power monitoring via SmartConnect Ethernet Port	4,201.00	6	25,206.12
Power Strips GL1&2 10 SL 2 HC 4 VC 1 SR 1	<u>Power Strip Tower Surge Protector, EyGde Electric Charging Station, 12 AC Outlets with 4 Individual ON/Off Switches, Flat Plug</u>	323.00	28	12,920.00

DR 1 NOC 3 ESL 5 MR 1 	<u>6ft Long Extension Cord with 5 USB Ports (1 USB-A Quick Charge, 4 USB-A)</u> • Features 12 AC outlets and 5 USB ports (including 1 Quick Charge USB-A and 4 standard USB-A ports), allowing simultaneous charging of multiple devices. • Tower design saves space and improves accessibility. • Provides surge protection to safeguard devices from power spikes.			
Cooling Units (Server and Database room) 	<u>Mitsubishi Electric MSZ-AP Series 7.1 kW Split System Air Conditioner MSZAP71VGD</u> • 7.1 kW cooling, suitable for medium to large spaces such as server rooms or open-plan areas. • Equipped with an anti-allergy enzyme filter to improve air quality. • Advanced inverter technology reduces energy consumption while maintaining consistent temperatures.	7,800.75	4	31,203.00
Cabling & Wiring				
Ethernet Cables (Cat6a)	<u>Cat6a 26AWG Snagless Shielded (S/FTP) PVC CM Ethernet Network Patch Cable</u>	13.30	1200m	15960

	• Up to 550MHz with 10G-T Speed, Support PoE/PoE+/PoE++ (IEEE 802.3af/at/bt) • S/FTP Cable Minimizes EMI and RFI Interference			
Fiber Optic Cables 	<u>Fiber Patch Cable, LC UPC to LC UPC, Single Mode (OS2)</u> • G.657.A1 Bend Insensitive Fiber & 10mm Min. Bend Radius • Precision Zirconia Ferrule Connectors Ensure Low Loss • OFNR Jacket for Vertical and Floor-to-floor Installations	20	200m	4000
Patch Panels 	<u>Cisco Catalyst 6500 WS-F6K-48X2-SPLTR</u> • Ports: 48 RJ45 Ports • Cable Compatibility: Cat6a, Cat6, Cat5e • Shielding: Shield • Build Quality: Industrial-grade, robust design • Ease of Installation: Designed for modular chassis, requiring specific configurations	268.80	5 (each lab installs 1)	1344
Keystone Jacks	ICC IC1078GAWH CAT6A RJ45 Keystone Jack	86	100	8600

	- Exceeds 500MHz transmission performance - Fully supports 10Gig Ethernet applications (10GBASE-T) - Compatible with CAT6, CAT5e, and voice network applications for versatility			
RJ45 Connectors 	Platinum Tools RJ45 SHIELDED CAT6 2PC CNNECTR - Eight conductor mod plug with loading bar - Uses 0.30mm (.012) 3 prong contact for solid conductors only	6.18	200	1236
Conduits and Trunking 	LINK - White uPVC Wiring Conduit Pipe (L19W) - Lightweight - Moisture Resistance - Smooth Inner Surface	1.7	210	357
Lab/Classroom/Office Devices				
Dekstop(GL, HC) 	CPU: Intel Core i5-13500 GPU: NVIDIA GeForce GTX 1650 Motherboard: ASUS Prime B760M-A DDR4 RAM: Corsair Vengeance DDR4 16GB (8GB x 2) Storage: Kingston NV2 1TB NVMe SSD	3660	93	340380

	• Power Supply Unit (PSU): Cooler Master MWE 550W • Case: Cooler Master MasterBox Q300L • Operating System: Windows 11 Pro (Licensed)			
Dekstop(CNL) 	• CPU: Intel Core i7-13700K • GPU: NVIDIA GeForce RTX 3060 Ti • Motherboard: ASUS ROG Strix Z690-A Gaming • RAM: Corsair Vengeance DDR4 32GB (16GB x 2) • Storage: Samsung 970 EVO Plus 1TB NVMe SSD • PSU: Seasonic Focus GX 750W • Case: NZXT H510 (~RM 350) • Operating System: Windows 11 Pro (Licensed)	7150	31	221650
Desktop(ESL) 	• CPU: Intel Core i7-13700KF • GPU: NVIDIA GeForce RTX 4060 Ti • Motherboard: MSI MPG Z690 Edge WiFi • RAM: Corsair Dominator Platinum 32GB (16GB x 2) DDR5 • Storage: WD Black SN850X 2TB NVMe SSD	8400	31	260400

	• PSU: Cooler Master V850 Gold • Case: Phanteks Eclipse P500A • Operating System: Windows 11 Pro (Licensed)			
Monitors 	<u>Dell 24 Monitor - P2422H</u> • Diagonal Size: 23.8 inches • Resolution: Full HD (1080p) 1920 x 1080 at 60Hz • Panel Technology: IPS • Adjustability: Height pivot(rotation) swivel tilt • Ports: DisplayPort VGA HDMI USB 3.2 Gen upstream 4 x USB 3.2 Gen 1 downstream Cables: 1x DisplayPort cable- DisplayPort to DisplayPort – 6ft, 1 x USB 3.2 Gen 1 upstream cable	642.25	140	89915
Smart TV (1 – CNL, 1 – ESL, 1 – HC) 	Samsung Neo QLED 8K QN800D Tizen OS Smart TV (2024) • Display: 75” • Refresh Rate: 100Hz • Resolution: 8K (7,680 x 4,320) • Video: NQ8 AI Gen2 Processor • AI Upscale: 8K AI Upscaling • Set Size with Stand (WxHxD): 1893.4 x 1153.4 x 303.7 mm	20799	3	62397

Projector (GL, Meeting Room) 	EPSON EB-X51 XGA 3800 Lumen 3LCD • Display Technology: 3LCD (RGB liquid crystal shutter projection system) • Brightness: White & Colour 3800 lumens • Resolution: XGA (1024 x 768) • Aspect Ratio: 4:3 • Contrast Ratio: 16,000:1 • Projection Size: 30" to 300"	1899	5	9495
Speaker (4 – 2 per GL 2 – HC 1 – CNL 1 – ESL 2 – MR) 	Yamaha NS-IC800 Atmos Ceiling Speaker (Pair) • Driver Size: 8-inch cone woofer with 1-inch tweeter • Frequency Response: 50Hz to 28 kHz • Sensitivity: 90 dB • Power Handling: 120W (recommended) • Impedance: 8 ohms	1099	10	10990
Microphone (1 per lab/classroom) 	Audio Technica ATR2100X-USB Cardioid Dynamic USB/XLR • Element: Dynamic • Polar Pattern: Cardioid • Frequency Response: 50-15k Hz • Bit Depth: Up to 24 bit	360	6	2160
Printers 	<u>HP LaserJet Pro M404dn</u> • Print Cartridges, Number: 1(black) • Print speed black (ISO, A4),Normal:	1250	7	8750

	Up to 38 ppm (default) ; Up to 40 ppm (HP High Speed) [1] • Processor speed: 1200 MHz • Media sizes: custom • Tray 1: 76 x 127 to 216 x 356 mm; Tray 2, optional Tray 3: 100 x 148 to 216 x 356 mm			
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Total Budget: RM 2,500,000.00

Total Cost: RM 1,438,595.17

Total Balance: RM 2,500,000.00 - RM 1,438,595.17 = RM 1,061,404.83

End Device: RM 1,006,137.00

Networking Device: RM 432,458.00

Balance: RM 2,500,000 - RM 432,458.00 = RM 2,067,541.83

3.3 Reflection

3.3.1 Are you surprised by the prices? How were you surprised?

After conducting an investigation, I was surprised to discover just how expensive network equipment can be. Even seemingly simple components, such as basic connectors, come with a high price tag, and many are needed in large quantities. Before this survey, I had assumed that most of the costs would be associated with large, sophisticated machines. However, I learned that building a comprehensive network system involves a multitude of devices and requires substantial funding. What struck me even more was the level of detail involved and the numerous factors that must be considered, such as selecting the right equipment for different systems and configurations.

3.3.2 Have you ever considered cost as a factor for choosing networking devices?

No, we prioritize the quality of all of the chosen devices to ensure that both devices meet all of the project's functional and performance demands. To make an informed decision, we compare the device's specifications, functions, and prices between the existing brands on the market, then we choose the most powerful and cost-effective one. If a device's cost is exceeding our budget, we seek for an alternative option which offer similar features and performance with a more affordable price.

3.3.3 What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.

1. Router

Choosing the right type of router is crucial for ensuring flexibility and scalability in a network. A router serves as the backbone of connectivity, and its capabilities directly impact how well a network can adapt to growing demands and evolving technologies. There are 2 brand of router that i find which is cisco ISR 4000 Series and Ubiquiti UniFi Dream Machine Pro where there has their own advantage but the cisco ISR 4000 Series meet our requirement.

Specifications	Cisco ISR 4000 Series	Ubiquiti UniFi Dream Machine Pro
Performance	Modular; up to 2 Gbps (depends on licenses and model)	Fixed; up to 3.5 Gbps (with IDS/IPS enabled)
Expandability	Highly expandable via modular design	Fixed hardware; integrates with other UniFi devices
WAN/LAN Ports	Modular (supports Ethernet, T1/E1, DSL, fiber modules)	Fixed (1x 10G SFP+ WAN, 8x 1G LAN ports)
Cost	RM 5,042.29	RM 1,698.02

The **Cisco ISR 4000 Series** stands out as the better choice for those prioritizing flexibility, scalability, and enterprise-grade features. Its modular design allows support for various WAN interfaces, including Ethernet, T1/E1, DSL, and fiber, making it highly adaptable to complex and evolving network requirements. Unlike the fixed hardware of the Ubiquiti UniFi Dream Machine Pro, the ISR 4000 enables organizations to customize their configurations to match specific needs, such as branch connectivity, WAN optimization, and SD-WAN capabilities. With advanced security features like DMVPN, FlexVPN, and zone-based firewalls, it provides robust protection for critical infrastructure. Additionally, its high scalability supports large networks with hundreds or thousands of users, making it ideal for this building where it need to support a large number of user simultaneous. While it comes at a higher cost, the Cisco ISR 4000 justifies its price with unmatched expandability, performance tailored to enterprise demands, and the ability to grow with your organization's needs.

2. Switches

When designing a network for a building with a large number of people, selecting the right type of switch becomes critical for ensuring both flexibility and scalability. In such environments, where the number of devices and users can change, a scalable switch allows the network to grow as needed without performance degradation. High-capacity switches with enough ports and fast uplink capabilities are essential to handle the large volume of data traffic generated by numerous devices.

Specifications	Ubiquiti UniFi Switch 8	Cisco Catalyst C9300L-48T-4X-E
Ports	8 Gigabit Ethernet ports	48 Gigabit Ethernet ports, 4x 10G/1G SFP+ uplinks
Manageability	Managed via UniFi Controller (cloud/local management)	Managed via Cisco IOS XE, highly customizable
Power over Ethernet (PoE)	No PoE support	PoE+ support, provides power to devices like IP phones, cameras, etc.
Uplink Ports	No dedicated uplinks, relies on the Gigabit Ethernet ports	4x 10G/1G SFP+ uplinks for high-speed connections to other switches or devices
Performance	Standard 1Gbps throughput	High-performance switch with enterprise-grade features, 10Gbps uplink capacity
Scalability	Limited scalability (8 ports only)	Highly scalable, designed for growing enterprise networks
Cost	RM 1,698.02	RM 9307.78

The **Cisco Catalyst C9300L-48T-4X-E** is the better choice when high-speed performance, scalability, and flexibility are critical. With 48 Gigabit Ethernet ports and 4x 10G/1G SFP+ uplinks, it offers superior speed and connectivity options, making it ideal for large, high-performance networks. The **10Gbps uplinks** ensure fast data transfer, which is essential for high-demand

applications, while the **48 ports** provide enough capacity to support a growing network. Unlike the **Ubiquiti UniFi Switch 8**, which is limited to only 8 ports, the **C9300L** is highly scalable, allowing it to easily adapt to expanding the needs, whether it's for adding more users, devices, or locations. Its flexibility is further enhanced by its **Cisco IOS XE** management, which offers advanced customization, security controls, and powerful network configurations, making it ideal for our networking. This combination of speed, scalability, and flexibility makes the C9300L-48T-4X-E an excellent choice due to its secure for the future network and support for a wide range of applications and expansion.

3. Cat6a Cable

Choosing the right Cat6a cable is critical to ensuring high-speed, reliable and future-proof network performance. It supports speeds of up to 10gbps and minimizes interference with improved shielding, making it ideal for data-heavy environments such as LABS and server rooms. Cat6a's durability and compliance with industry standards provides a cost-effective solution that meets today's needs while allowing for future scalability, ensuring that the network remains efficient and robust over time.

Specifications	Cat6a 26AWG Snagless Shielded (S/FTP)	Belden Cat6A Cable
Durability	Moderate (suitable for indoor use)	High (with LSZH for harsh environments)
Flexibility	High (Snagless design prevents cable damage)	Moderate
AWG (Wire Gauge)	26AWG	23AWG
Cost	RM13.30/m	RM47.60/m

The Cat6a 26AWG Snagless Shielded (S/FTP) cable is the better choice due to its excellent balance of performance, flexibility, and cost-efficiency. With a snagless design, it offers high flexibility, making installation and management easier, especially in confined spaces or complex setups. While its durability is moderate and suitable for indoor environments, it is adequate for most use cases within your building. Additionally, its cost of RM13.30/m is significantly more affordable compared to the Belden Cat6A cable's RM47.60/m, delivering substantial savings without compromising on essential features like 10 Gbps speed and shielding against interference. For indoor applications with moderate interference, this cable provides all necessary functionality at a fraction of the cost.

4. Fiber Optic Cable

Choosing the right Cat6a cable is critical to ensuring high-speed, reliable and future-proof network performance. It supports speeds of up to 10gbps and minimizes interference with improved shielding, making it ideal for data-heavy environments such as LABS and server rooms. Cat6a's durability and compliance with industry standards provides a cost-effective solution that meets today's needs while allowing for future scalability, ensuring that the network remains efficient and robust over time.

Specifications	Fiber Patch Cable, LC UPC to LC UPC, Single Mode (OS2)	Fiber Optic Jumper, 1 F, ST
Fiber Type	Single Mode (OS2)	Single Mode
Connector Type	LC UPC to LC UPC	ST Compatible to ST Compatible
Insertion Loss	≤ 0.3 dB	≤ 0.4 dB
Cost	RM20/m	RM30/m

The Fiber Patch Cable, LC UPC to LC UPC, Single Mode (OS2) is the better choice due to its superior cost-effectiveness and compatibility with modern networking equipment. It offers a lower cost of RM20/m compared to the Fiber Optic Jumper's RM30/m, providing significant savings without sacrificing performance. Additionally, the LC UPC connectors are more compact and widely used in high-density networking setups, making them ideal for modern switches and equipment. With a lower insertion loss of ≤ 0.3 dB, it ensures better signal quality and efficiency over long distances, while the ST-compatible jumper's ≤ 0.4 dB is slightly less efficient. This makes the LC UPC patch cable a more practical and economical option for your networking needs.

5. PC (Personal Computer)

For PC, a powerful processor allows a PC perform multitasking and productivity tasks smoothly and effectively. For the important lab like CNL and ESL, Intel Core i7-13700K processor will be used in the PC for each workstation. The alternative one is AMD Ryzen 7 7800X3D.

Specification	Inter Core i7-13700K	AMD Ryzen 7 7800X3D
Cores/Threads	16/24	8/16
Base Clock Speed	3.4 GHz	4.2 GHz
Max Boost Clock Speed	5.4 GHz	5.0 GHz
L3 Cache	30 MB	96 MB
Integrated Graphics	Yes	No
Price (Approx.)	RM 1580.00	RM 2500.00

The Intel Core i7-13700K provides more cores and threads, which can enhance better multitasking and productivity tasks, which is essential for academic labs to handle multiple simulations and design task simultaneously. Moreover, it includes integrated graphic which provides flexibility if a discrete GPU isn't available. Even though AMD Ryzen 7 7800X3D has larger L3 cache and base speed, but considering overall versatility and performance, Intel Core i7-13700K will be a more efficient choice.

6. Smart TV

A Smart TV is essential to display high-quality video conferencing and multimedia presentations. For our plan, we will use Samsung QN800D Neo QLED 8K, while the alternative one will be LG Z3 8K OLED TV.

Specification	Samsung QN800D Neo QLED 8K	Sony BRAVIA 9 85-inch Mini LED QLED 4K Ultra HD 120Hz Smart TV (Google TV) 2024
Display Technology	Neo QLED (Mini LED)	Mini LED QLED
Resolution	8K (7680x4320)	4K Ultra HD (3840x2160)
HDR Support	HDR10+, HLG	HDR10, Dolby Vision, HLG
Operation System (OS)	Tizen OS	Google TV
Refresh Rate	120Hz	120Hz
Audio	70W 4.2.2 channel with Dolby Atmos	Multi-directional speakers with Dolby Atmos
Price (Approx.)	RM 20799.00	RM 24399.00

Samsung QN800D Neo QLED 8K has a higher resolution which is 8K, which provides superior clarity for detailed technical presentations which is beneficial in lab and classroom. Moreover, it is equipped with a 70W 4.2.2 channel sound system featuring built-in up, down, and side-firing speakers with True Dolby Atmos support which can deliver immersive audio experiences which enhance engagement during lectures.

7. Projector

A projector is crucial to display presentation, and collaborative contents. The Epson EB-X51 XGA 3800 Lumens 3LCD Projector will be used due to its excellent resolution and brightness. The alternative one is BenQ MS560 SVGA 4000 Lumens Projector.

Specification	EB-X51 XGA 3800 Lumens 3LCD Projector	BenQ MS560 SVGA 4000 Lumens Projector.
Resolution	XGA (1024x768)	SVGA (800x600)
Brightness	3800 lumens	4000 lumens
Price (Approx.)	RM 1899.00	RM 1499.00

Comparing to BenQ MS560 SVGA 4000 Lumens Projector, EB-X51 XGA 3800 Lumens 3LCD Projector has higher native resolution, which provides sharper image quality. This is essential for academic and professional presentations. Although it is more expensive than the BenQ MS560 SVGA 4000 Lumens Projector, its specification can meet the demands for the hybrid classroom and general-purpose labs.

8. Speaker

High-quality speakers needed for each classroom and lab in the building. In our plan, Yamaha NS-IC800 Atmos Ceiling Speaker (Pair) will be installed in each classroom and labs, while Polk Audio RC80i Ceiling Atmos speakers will be the alternative,

Specification	Yamaha NS-IC800 Atmos Ceiling Speaker (Pair)	Polk Audio RC80i Ceiling Atmos speakers
Speaker Type	In-ceiling, Atmos-ready	In-ceiling, Atmos-ready
Frequency Response	50Hz – 28kHz	35Hz – 20kHz
Sensitivity	90 dB	90 dB
Power Handling	140W (Max)	100W (Max)
Price (Approx.)	RM 1099.00	RM 1299.00

Compare to Polk Audio RC80i, Yamaha NS-IC800 has a wider frequency range (50Hz – 29kHz), which allow it to capture finer audio details which is beneficial for technical presentations and lectures. Furthermore, Yamaha NS-IC800 has a higher maximum power capacity of 140W, which suitable for larger spaces. Also considering the price, Yamaha NS-IC800 will be more cost-effective while provide better performance.

9. Microphone

A reliable and high-quality microphone is important for each lab and classroom to ensure clear and sharp audio for video conferencing and lectures. In our plan, we will use Audio Technica ATR2100X-USB Cardioid Dynamic USB/XLR, the alternative one will be Samson Q2U Handheld Dynamic USB.

Specification	Audio Technica ATR2100X-USB Cardioid Dynamic USB/XLR	Samson Q2U Handheld Dynamic USB
Polar Pattern	Cardiod	Cardiod
Frequency Response	50Hz – 15kHz	50Hz – 15kHz
Connectivity	USB-C, XLR	USB, XLR
Bit Depth/Sample Rate	24-bit/192kHz	16-bit/48kHz
Price (Approx.)	RM 360.00	RM 390.99

Both microphone has similar price, but Audio Technica ATR2100X-USB supports 24-bit/192kHz which provide superior audio clarity and precision for live communication, which is important for technical explanation and presentation. Moreover, it includes USB-C which ensures better compatibility with modern devices. Therefore, with a lower cost, Audio Technica ATR2100X-USB will be our first choice.

10. Patch Panel

A high-quality patch panel is crucial to ensure the efficiency of cable management and network connectivity. The Cisco Catalyst 6500 WS-F6K-48X2-SPLTR is chosen in our plan on count of its

excellent compatibility. While the alternative choice is TRENDnet 48-Port Cat6 Unshielded Patch Panel (TC-P48C6U).

Specification	Cisco Catalyst 6500 WS-F6K-48X2-SPLTR	TRENDnet 48-Port Cat6 Unshielded Patch Panel (TC-P48C6U)
Ports	48 Ports	48 Ports
Cable Compatibility	Cat6a, Cat6, Cat5e and fiber cable	Cat6, Cat5e
Shielding	Unshielded	Unshielded
Build Quality	Durable metal frame with high-quality construction for enterprise-level networks	Solid build for typical network installations
Management Features	No additional cable management bar, but offers more flexibility for network environments.	Comes with a cable management bar included for better organization.

Both Cisco Catalyst 6500 WS-F6K-48X2-SPLTR and TRENDnet 48-Port Cat6 Unshielded Patch Panel (TC-P48C6U) provide the same number of ports for high-density installations, but Cisco Catalyst 6500 provides better overall compatibility with fiber cables which allow it to connect with the fiber optic cabling in our plan. This can enhance flexibility as the network grows especially in the data-heavy environments like the CSL and ESL. As the result, Cisco Catalyst 6500 WS-F6K-48X2-SPLTR become our first choice due to its lower price and better build quality which is able to meet the networking demands in the building.

11. Access Points (Wi-Fi)

Reliable and convenient access points are crucial as it is the backbone of any computing building's digital infrastructure and wireless networking by providing internet access to wireless devices. In our plan, we decided to use the C9130AXI-A, while the other counterpart choice is the TP-Link EAP225.

Specifications	Cisco Catalyst C9130AXI-A	TP-Link EAP225
Wi-Fi Standard	Wi-Fi 6 (802.11ax)	Wi-Fi 5 (802.11ac)
Max Data Rate	5.38 Gbps (dual 5 GHz, 2.4 GHz combined)	1.75 Gbps (1300 Mbps on 5 GHz, 450 Mbps on 2.4 GHz)
Ethernet Ports	1 x 1 GbE (supports PoE)	1 x Gigabit Ethernet
Security Features	WPA3, WPA2, 802.1X, AES, EAP	WPA3, WPA2, WPA, WEP
Coverage Area	Larger scale (enterprise-level)	Small to medium-scale (SMB)

The C9130AXI-A is designed for large, high-demand environments, offering more robust features for dense deployments, while the TP-Link EAP225 is more suitable for smaller businesses or home networks with fewer clients and less traffic. As the final conclusion, the C9130AXI-A is chosen for its large-scale enterprise deployment as it offers several advanced features aimed at providing robust

wireless coverage and high-speed connections making it suitable for environments with dense device populations.

DATE/TIME	16 Nov 2024 10:00 pm		
LOCATION	M01 KTDI + Online		
AGENDA	1. Understand the details about task 3		
	2. Discuss the materials prepared in advance		
	3. Determine the possible network devices to be included in project and the quantity and the price for them		
	4. Task distribution for all members		
Meeting MC	LEE YIN SHEN		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
TAN ZHI MING A23CS0189	22:05		
NG YU HIN A23CS0148	22:00		
LEE YIN SHEN A23CS0236	22:00		
MUHAMMAD NAIM BIN ABDULLAH A23CS0134	22:05		
MINUTES			
NO.	ITEM DISCUSSED	IDEA/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion for task 3	- Tan Zhi Ming shared the rubric and the information of the task - All members read and investigate the task carefully - Clearly identify what is the objective of the task	Yu Hin
2	Suggestion for network devices that will be included in project	- Yin Shen and Naim suggest the network devices that will be used in ground floor - Zhi Ming and Yu Hin suggested the network devices that will be used in first floor	All members
3	Task Distribution	• Muhammad Naim – Devices of Power and cooling system • Tan Zhi Ming – Devices of core system and cabling • Ng Yu Hin – Devices of each lab • Lee Yin Shen – Devices of office and classroom The first question was assigned to Tan Zhi Ming and the second question assigned to Lee Yin Shen The comparative question assigned to all of	All members

		member.	
5	Meeting ended	0:00	

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